$32+16+8+0+2+1 = (59)_{10}$

$(111011)_2 = (59)_{10}$

2.4.6 Binary to Octal Conversion

Step 1: Group the given binary number into 3 bits from right to left.

Step 2: You can add preceding 0 to make a group of 3 bits if the left most group has less than 3 bits.

Step 3: Convert equivalent octal value using "2's power positional weight method"

Table 2.4 Octal numbers and their Binary equivalent

Octal	Binary Equivalent
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Example

Convert $(11010110)_2$ into octal equivalent number

Step 1: Group the given number into 3 bits from right to left.

011 010 110

Note: The left most groups have less than 3 bits, so 0 is added to its left to make a group of 3 bits.

Step-2: Find Octal equivalent of each group

$$\underbrace{011}_{3}\quad\underbrace{010}_{2}\quad\underbrace{110}_{6}$$

$(11010110)_2 = (326)_8$

2.4.7. Binary to Hexadecimal Conversion

Step 1: Group the given number into 4 bits from right to left.

Step 2: You can add preceding 0's to make a group of 4 bits if the left most group has less than 4 bits.

Step 3: Convert equivalent Hexadecimal value using "2's power positional weight method"

Example

Convert $(1111010110)_2$ into Hexadecimal number

Step 1: Group the given number into 4 bits from right to left.

0011 1101 0110

Note: 0's are added to the left most group to make it a group of 4 bits

$$\underbrace{0011}_{3}\quad\underbrace{1101}_{D}\quad\underbrace{0110}_{6}$$

$(1111010110)_2 = (3D6)_{16}$

2.4.8 Conversion of fractional Binary to Decimal equivalent

Follow the steps to convert fractional Binary number to its Decimal equivalent.

Step 1: Convert integral part of Binary to Decimal equivalent using positional notation method (Procedure is same as discussed in 2.4.5)

Step 2: To convert the fractional part of binary to its decimal equivalent.

Step 2.1: Write down the Binary digits in the fractional part

Step 2.2: For all the digits write powers of 2 from left to right starting from $2^{-1}, 2^{-2}, 2^{-3}, \ldots 2^{-n}$, now write the equivalent weight.

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